

Task 1 Screenshots

Task 1: Data Cleaning & Preprocessing

```
[1]: import pandas as pd
import numpy as np
```

```
[3]: df = pd.read_csv("Titanic-Dataset.csv")
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

```
[4]: df.isnull().sum()
```

```
[4]: PassengerId    0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

Handle Missing Values

```
[7]: age_median = df['Age'].median()
df['Age'] = df['Age'].fillna(age_median)
```

```
[8]: df['Cabin'] = df['Cabin'].fillna('Unknown')
```

```
[9]: embarked_mode = df['Embarked'].mode()[0]
df['Embarked'] = df['Embarked'].fillna(embarked_mode)
```

Convert Datatypes

```
[11]: df['Age'] = df['Age'].astype(int)
```

```
[12]: categorical_cols = ['Survived', 'Pclass', 'Sex', 'Embarked']
for col in categorical_cols:
    df[col] = df[col].astype('category')
```

Rename Columns

```
[13]: rename_map = {
    'PassengerId': 'passenger_id',
    'Survived': 'is_survived',
    'Pclass': 'passenger_class',
    'Name': 'passenger_name',
    'Sex': 'gender',
    'Age': 'age',
    'SibSp': 'siblings_spouses_count',
    'Parch': 'parents_children_count',
    'Ticket': 'ticket_number',
    'Fare': 'fare_paid',
    'Cabin': 'cabin_code',
    'Embarked': 'embarkation_port'
}

df = df.rename(columns=rename_map)
```

Cleaned Dataset Summary